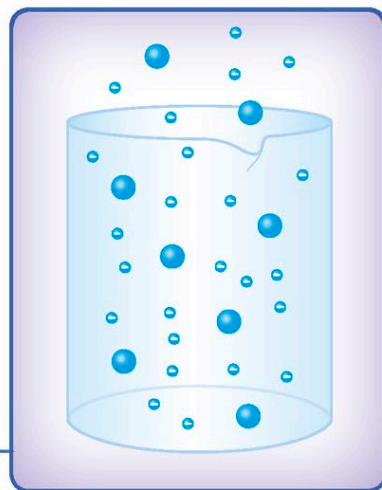
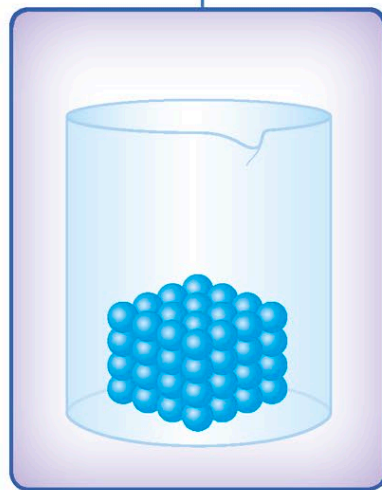




Plasma
Plasma, which makes up the sun and stars, is the most common matter in the universe. Intense heat makes its atoms separate into positively charged nuclei and negatively charged electrons that whiz about at very high speed.

Aurora borealis
Collisions between plasma from space and gases in the atmosphere energize atmospheric atoms, which release light when they return to normal energy levels.

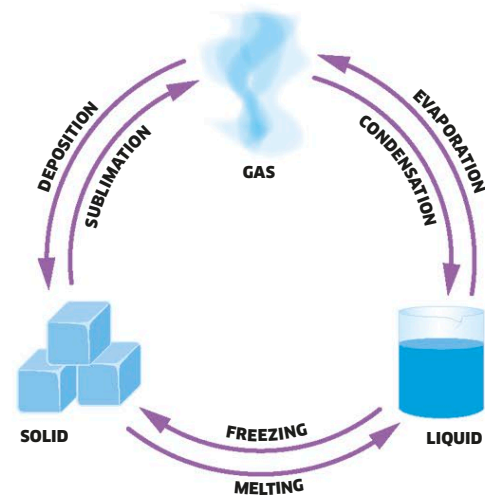
States of matter
Water exists in three states. Here we see it as solid ice, liquid seawater, and gaseous water vapor exhaled by the polar bear. Water vapor is invisible until it cools and condenses to form steam, a mist of liquid droplets—the same happens when a pan of water boils. In the Arctic Circle, the spectacular northern lights (aurora borealis) reveal the presence of plasma, the fourth state of matter.



Solid state
In a solid, such as ice, particles are held together by bonds and sit tightly packed. The particles vibrate slightly but they don't move around, so solids keep their shape.

Changing states of matter

Adding or removing energy (as heat) causes a state change. Solids melt into liquids, and liquids vaporize into gas. Some solids can turn straight to gas; some gases into solids.



Sublimation
Solid carbon dioxide is known as dry ice. With lowered pressure and increased heat it becomes CO₂ gas—this is called sublimation. When a gas goes straight to solid, the term is deposition.

Melting and freezing

All pure substances have a specific melting and freezing point. How high or low depends on how their molecules are arranged.

